Sheetname: Power Management

Sets 5V and 3V3

Sheetfile: pwr_mgmt.kicad_sch

Sheetname: ESP Reciever

BCK
DIN
LRCLK

File: esp_rx.kicad_sch

Sheetname: DAC

BCK
DIN
LRCLK
DAC_OUT_L
DAC_OUT_R

File: dac.kicad_sch

Sheetname: Volume Control

DAC_OUT_L
DAC_OUT_R
POST_POT_IN_L
POST_POT_IN_R
SHDN_SIGNAL

File: vol_ctrl.kicad_sch

Sheetname: Amp

POST_POT_IN_L
POST_POT_IN_R
SHDN_SIGNAL
AMP_OUT_L
AMP_OUT_R

File: amp.kicad_sch

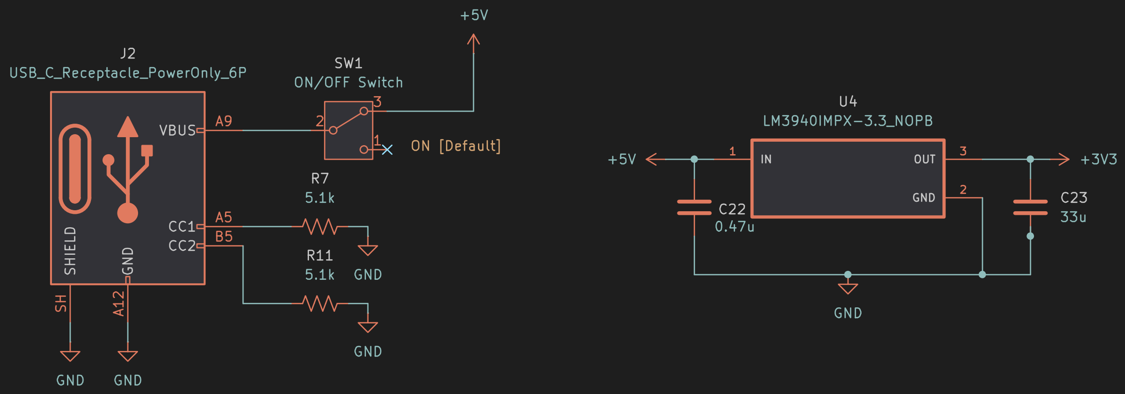
Dual Gang pot has a hard mute at 0% volume.
DAC features a soft mute switch.
Data: ESP -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /
File: ESP32IEMCAD.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

Size: A5	Date: 2026-08-04	**Rev: 0.0.1**
KiCad E.D.A. 10.0.5		Id: 1/6



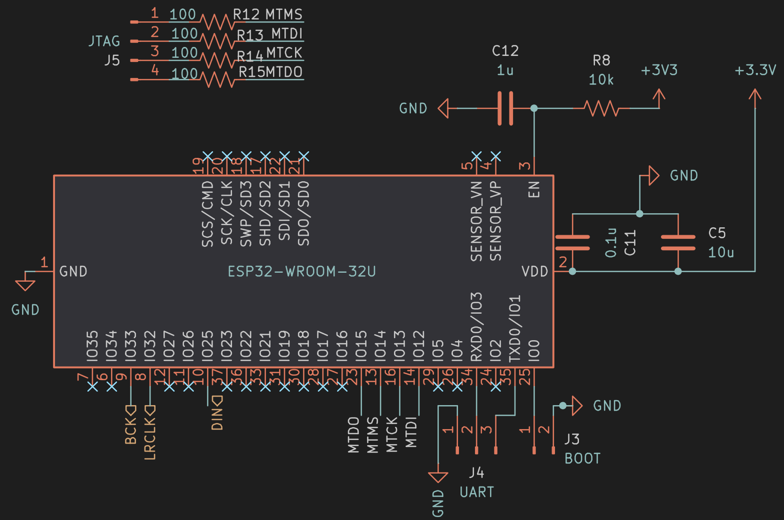
Using a low dropout regulator to step from 5V to 3V3 for ESP, DAC, and Amp.
CC1, CC2 have a pull down resistor to GND to signify it is an Upstream Facing Port (UFP)
Connect to a USB-C power bank. Will provide power to the entire RX circuit.
Data: ESP -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /Power Management/
File: pwr_mgmt.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

Size: A5	Date: 2026-08-04	**Rev: 0.0.1**
KiCad E.D.A. 10.0.5		Id: 2/6



Recieves the compressed data from ESP TX and decodes them. Then sends to DAC.

Data: ESP -> DAC -> AMP-> 3.5mm Jack.

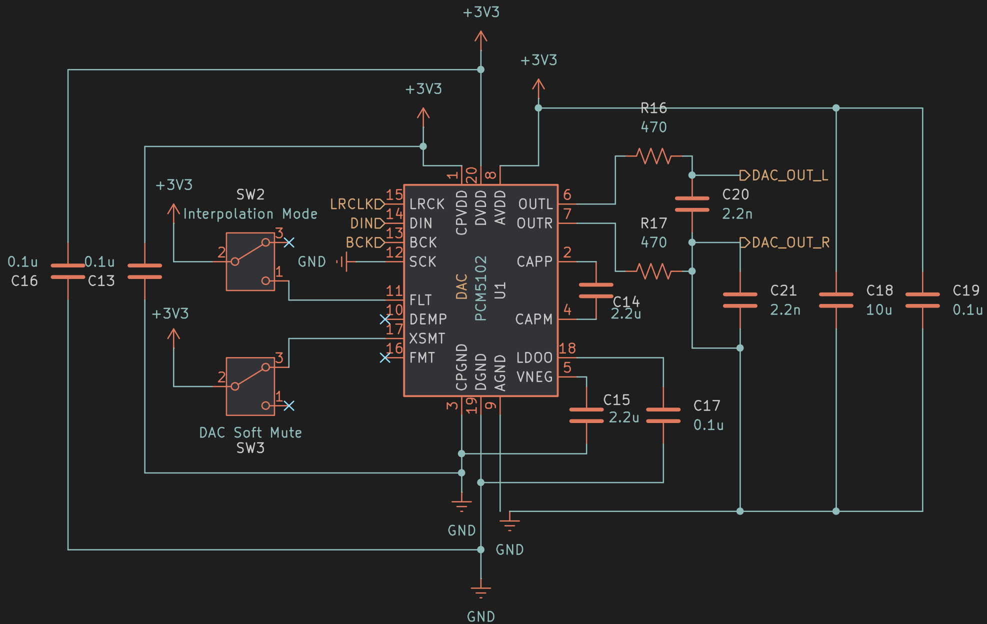
damian m

Sheet: /ESP Reciever/

File: esp_rx.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

Size: A5	Date: 2026-08-04	**Rev: 0.0.1**
KiCad E.D.A. 10.0.5		Id: 3/6



Use PLL functional diagram since we are grounding SCK

DEMP Low = DISABLE De-emphasis eq (44.1kHz).
High = ENABLE De-emphasis eq (44.1kHz).

FMT Low = I2S Format.
High = Left justified.

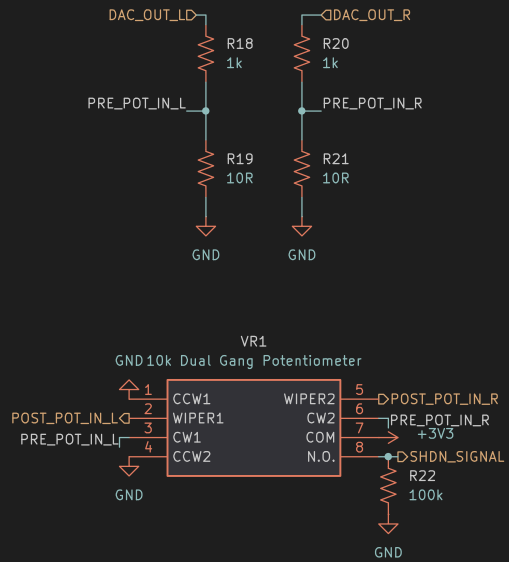
Soft Mute: XSMT LOW = Muted. XSMT HIGH = Unmuted. [Default]
Interpolation mode: FLT HIGH = Low Latency. FLT LOW = Normal Latency [Default]
Data: ESP -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /DAC/
File: dac.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

Size: A5	Date: 2026-08-04	**Rev: 0.0.1**
KiCad E.D.A. 10.0.5		Id: 4/6



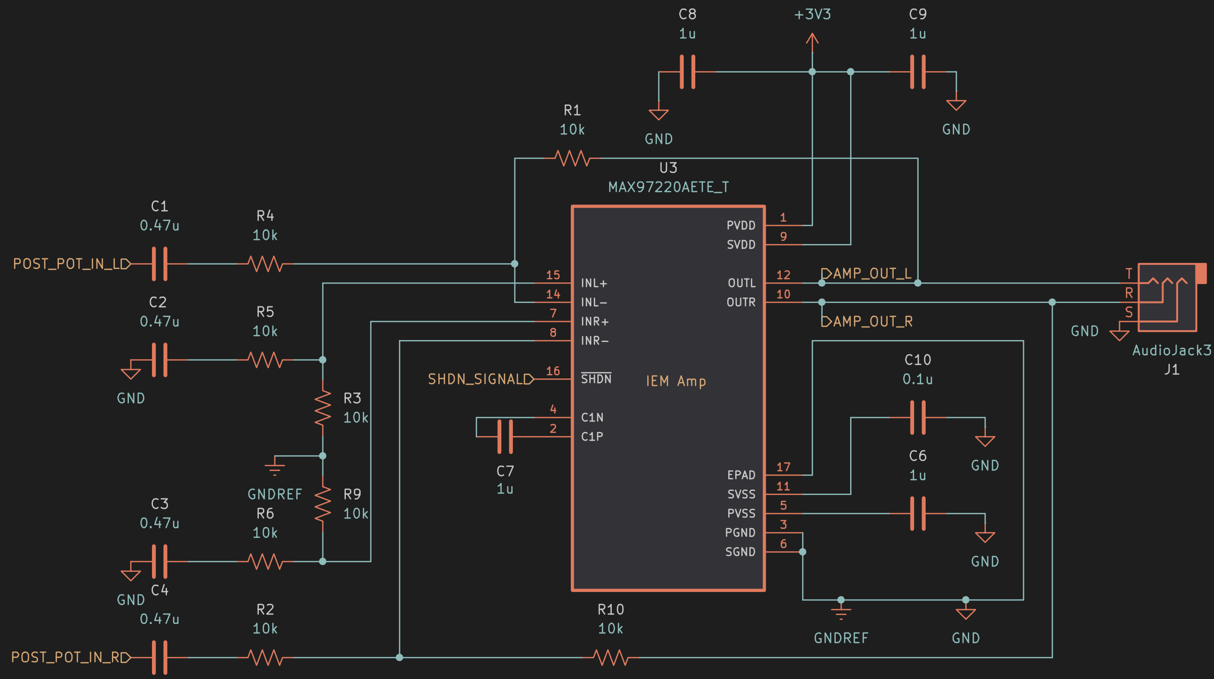
POST_POT_IN_R/L becomes the input to the amp.
Volume Knob with integrated off switch to kill amp from playing any audio.
AMP_IN_L/R follows a voltage divdier to lower volume to safer levels before hitting AMP.
Data: ESP -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /Volume Control/
File: vol_ctrl.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

Size: A5	Date: 2026-08-04	**Rev: 0.0.1**
KiCad E.D.A. 10.0.5		Id: 5/6



SHDN_SIGNAL LOW acts as setting the volume to a "true 0%"
by shutting off the IEM amp.

This is in contrast to the DAC Soft mute which acts as a
mute toggle without
changing the volume.

Amp follows the pseudo-differential input configuration
because the DAC output is a single-ended input.

SHDN_SIGNAL is handled by the dual gang pot. N.O. = AMP OFF.
BIAS is in the same position as SVSS. Refer to in manual as BIAS.
Data: ESP -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /Amp/
File: amp.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

Size: A5	Date: 2026-08-04	**Rev: 0.0.1**
KiCad E.D.A. 10.0.5		Id: 6/6